

Project Documentation

Date	Team ID	Project Name
30 September 2026	SWTID-2026-9981	PocketSmart AI: Your Smart Budget & Recommendation Assistant

Team Details

S.No	Name	Role
1	K Devishree	Team Leader
2	Pon Sujatha M	Team Member
3	Karpagam N	Team Member
4	Deepika M	Team Member
5	Brindha K	Team Member

Section	Description
1. Introduction	PocketSmart AI is a web-based smart budget and recommendation assistant. It helps users record expenses, understand spending through automatic categorisation and charts, stay within category budgets through alerts and receive personalised saving tips and budget-friendly suggestions generated with AI.
2. Problem Statement	People with limited monthly income struggle to track expenses, notice overspending in time and make sensible buying decisions. Existing tools are mostly manual and give little personalised guidance.
3. Objectives	Record expenses easily; categorise them automatically; warn users at 80 percent and 100 percent of each category budget; visualise spending; give personalised recommendations within the remaining budget; answer finance questions in plain language; provide a downloadable monthly report.
4. Scope	In scope: registration and profile, expense management with CSV import, automatic categorisation, budgets and alerts, dashboard, AI recommendations, chat assistant, CSV export, online deployment. Out of scope for this version: automatic bank synchronisation, bill reminders and native mobile app.
5. System Requirements	Hardware: any computer or smartphone with a web browser and internet. Software: Python 3, Streamlit, pandas, scikit-learn, Plotly, SQLite, Gemini API key, Git and GitHub.
6. Architecture and Design	Layered design with a Streamlit interface, a Python application layer (authentication, expenses, categoriser, budget engine, recommendation engine, chat), SQLite storage, a product catalogue and the Gemini API. See the Design Phase folder for the problem-solution fit, proposed solution and architecture diagram.
7. Implementation	Development followed three sprints: Sprint-1 (profile, login, expenses), Sprint-2 (categorisation, budgets, dashboard, CSV import), Sprint-3 (recommendations, chat, export, deployment). Code is organised into modules for authentication, expenses, categoriser, budget and recommendations.
8. Testing	17 functional test cases cover registration, expenses, categorisation, budgets, charts, import, recommendations, chat, export and deployment. Performance targets cover load time, response time, import speed and concurrent usage.
9. Results and Benefits	Users can see spending clearly, receive early warnings and get relevant suggestions, which supports better saving habits without sharing bank credentials.

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10. Limitations	Expenses must be entered or uploaded by the user; categorisation accuracy depends on the training data; AI suggestions depend on the availability of the LLM service and the catalogue.
11. Future Scope	Automatic bank statement import, bill and subscription reminders, savings goal planner with timelines, regional language support, voice input and a mobile application.
12. Conclusion	PocketSmart AI shows how machine learning and a large language model can be combined in a simple application to turn everyday expense data into clear, personalised financial guidance.

Team Contribution

Team Member	Role	Contribution
K Devishree	Team Leader	Coordination, architecture, LLM integration, recommendations, deployment
Pon Sujatha M	Team Member	User interface, dashboard charts, profile screens, report export
Karpagam N	Team Member	Training data, categorisation model, CSV import
Deepika M	Team Member	Database, registration, login, expense management
Brindha K	Team Member	Budget and alert engine, chat assistant, testing, documentation